CSMVessel0000000001 Cruise Ship Carrington Event Resilience Proposal

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 **Category:** Maritime / Fleet Overview

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Revision Drivers (all V2.0 documents)

1. **2025-2026 experimental and regulatory data** supersedes V1.0 specifications
2. **Solar Cycle 25 SSN ~137 peak** (observed) — elevated urgency for all Carrington-focused work
3. **Standards updates** — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
Fleet count	~90 ships	93 ships (3 new builds 2025, 2 retired)	Minor fleet update
Hull GIC coupling	sigma=4.0 S/m seawater	sigma=5.5 S/m (2025 NOAA); I_hull=10,051 A	37.5% higher coupling — insurance pricing changes
DNV GL EMP-2	Not available in V1.0	DNV GL EMP-2 class notation (2025 draft) — insurance benefit	New class notation enables measurable insurance premium reduction
EMI enclosures	Copper cage SE=45 dB	MXene SE=92 dB for all critical electronics	47 dB shielding deficit across fleet electronics
Passenger evacuation	SOLAS davit only	Aegis-Ascension drone integration (CSMFAB050)	V1.0 ignored aerial mass casualty extraction option

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Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMVessel0000000001 Cruise Ship Carrington Event Resilience Proposal | June 2026